

```

10.
11.  # Initialize averages with simple averages of initial values
12.  avg_gain = sum(gains[:period]) / period
13.  avg_loss = sum(losses[:period]) / period
14.
15.  # Use standard RSI smoothing formula for remaining values
16.  for i in range(period, len(changes)):
17.      avg_gain = (avg_gain * (period - 1) + gains[i]) / period
18.      avg_loss = (avg_loss * (period - 1) + losses[i]) / period
19.
20.  # Calculate RS and RSI
21.  if avg_loss == 0:
22.      return 100 # No losses means maximum bullish signal
23.
24.  rs = avg_gain / avg_loss
25.  rsi = 100 - (100 / (1 + rs))
26.
27.  return rsi
28.

```

This function calculates RSI using the standard Wilder's formula, which professional traders have relied on for decades.

Signal Generation Functions

Signal generation functions use indicators to identify potential trading opportunities:

```

1. def check_rsi_oversold(prices, threshold=30, period=14):
2.     # Calculate RSI
3.     rsi_value = calculate_rsi(prices, period)
4.
5.     # Check if oversold (RSI below threshold)
6.     is_oversold = rsi_value < threshold
7.
8.     return is_oversold, rsi_value # Return both the signal and the value
9.

```

This function identifies potential buying opportunities when RSI drops below an oversold threshold.